Clément Basdevant

Python & Data Software Engineer

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SUMMARY

Python & data software engineer, SeaTech graduate (Grenoble INP): web and desktop interfaces (Vue.js, PyQt), data processing, time series forecasting and deep learning (PyTorch). Seeking a software or data role in Grenoble.

SKILLS

Languages: Python, C++, SQL, JavaScript, Matlab
Data & AI: pandas, NumPy, SciPy, PyTorch, N-BEATS, machine learning, deep learning, time series, signal processing
Web & UI: Vue.js, Quasar, PyQt / Qt, HMI, UI/UX
Data & systems: PostgreSQL, TimescaleDB, Kafka, REST API, Docker, Linux, DDS (CycloneDDS), MPI
Methods & quality: Git, Agile / Scrum, property-based testing (Hypothesis), mutation testing, CI (GitHub Actions), PyPI packaging
GenAI & agents: Claude Code, AI agents, prompt engineering
Spoken languages: French (native), English B2/C1 (TOEIC 880), German B1

PROFESSIONAL EXPERIENCE

Research Engineer, Software & Data – G2Elab, Grenoble, France **Jun 2025 – Sep 2026**
xKy start-up project: a sensor that streams Linky smart-meter data in real time

- Built, as a team, an infrastructure able to scale beyond 10,000 sensors.
- Developed a Vue.js / Quasar web application, deployed with Docker, for end users and researchers.
- Electricity consumption forecasting (N-BEATS, PyTorch) and analysis, visualisation and decision-support tools.
- Co-lead of the CaStore project (predictive analytics for energy data): data, forecasting and UX.
- Start-up creation: market research, go-to-market strategy, Pollutec 2025 trade show; Agile team of 4.

Test Software Development Engineer (Master's internship) – Trixell, Moirans, France **Mar 2024 – Aug 2024**
Digital X-ray detectors for medical imaging · Test team, Software and Image group

- Developed Python tools communicating with the detectors over the DDS protocol (CycloneDDS).
- Designed an HMI to explore DDS traffic, integrated into the team's test framework (IEC 62304, medical devices).

Python Developer (international internship, in English) – NTNU, Trondheim, Norway **May 2023 – Jul 2023**

- Built a new PyQt interface for REEF3D (hydrodynamic simulation), with guided input of the software's 586 parameters.
- Launched parallel computations (MPI) from the interface; ported it to Windows, Linux and macOS.

Python Developer (first-year internship) – CORYS, Grenoble, France **Jun 2022 – Jul 2022**

- Converted 5 Excel calculation spreadsheets into Python tools (pandas, Matplotlib) with graphical interfaces.
- Replaced the Excel solver with SciPy optimisation (BFGS); systematic validation of the results.

OPEN-SOURCE PROJECT

dsp-props, property-based testing library for Python – author, open source (BSD-3) **2026**

- Checks the mathematical properties of signal-processing code (linearity, Parseval…) with Hypothesis and minimal counterexamples; on PyPI: github.com/clement-basdevant/dsp-props.
- **Contribution merged into SciPy 2.0.0**: fix for scipy.signal.lfilter and sosfilt on empty inputs (PR #26415), from a bug found with dsp-props.

OTHER EXPERIENCE

Trainee Cross-Country Ski Instructor (self-employed) – École de Porte, Col de Porte, France **Since Dec 2025**

- Cross-country skiing and biathlon lessons, mostly for children; coach at the Ski Nordique Chartreuse club.

EDUCATION

Master's degree in Engineering (Diplôme d'ingénieur) – SeaTech, Grenoble INP partner school, Toulon **2021 – 2024**

- IRIS track (data science, information, systems); deep learning project in computer vision.

Preparatory classes for engineering schools (CPGE PTSI) – Lycée Vaucanson, Grenoble **2019 – 2021**

CERTIFICATIONS & AWARDS

- **Prix Lamblin 2024** (student award, SeaTech) | Amateur radio licence | State ski instructor diploma (in progress)

INTERESTS

- Competitive running (CAPRG club): 10 km road in 33'18, 5 km road in 15'00, 1,500 m in 3'59"66.
- Paragliding (FFVL pilot licence); volunteer at the Coupe Icare festival. Secretary of the SeaTech Alumni association.